As the average investor becomes more risk averse to variance (so $\bar{\gamma}$ increases), the risk premium of the market also increases.

3.5. CAPM LESSON 5: RISK IS FACTOR EXPOSURE

The risk of an individual asset is measured in terms of the factor exposure of that asset. If a factor has a positive risk premium, then the higher the exposure to that factor, the higher the expected return of that asset.

The second pricing relationship from the CAPM is the traditional beta pricing relationship, which is formally called the security market line (SML). Denoting stock i's return as r_i and the risk-free return as r_f, the SML states that any stock's risk premium is proportional to the market risk premium:

$$\begin{aligned} E(r_i) - r_f &= \frac{\operatorname{cov}(r_i, r_m)}{\operatorname{var}(r_m)} \left(E(r_m) - r_f\right) \\ &= \beta_i \left(E(r_m) - r_f\right). \end{aligned} \tag{6.2}$$

The risk premium on an individual stock, $E(r_i) - r_f$, is a function of that stock's beta, $\beta_i = \operatorname{cov}(r_i, r_m)/\operatorname{var}(r_m)$. The beta is a measure of how that stock co-moves with the market portfolio, and the higher the co-movement (the higher $\operatorname{cov}(r_i, r_m)$), the higher the asset's beta.

I will not formally derive equation (6.2).[4] But it contains some nice intuition: mean-variance investing is all about diversification benefits. Beta—the CAPM's measure of risk—is a measure of *the lack of* diversification potential. Beta can be written as $\beta_i = \rho_{i,m}\sigma_i/\sigma_m$, where $\rho_{i,m}$ is the correlation between asset i's return and the market return, σ_i is the volatility of the return of asset i, and σ_m is the volatility of the market factor. Chapter 3 emphasized that the lower the correlation with a portfolio, the greater the diversification benefit with respect to that portfolio because the asset was more likely to have high returns when the portfolio did badly. Thus, *high* betas mean *low* diversification benefits.

If we start from a diversified portfolio, investors find assets with high betas—those assets that tend to go up when the market goes up, and vice versa—to be unattractive. These high beta assets act like the diversified portfolio the investor already holds, and so they require high expected returns to be held by investors. In contrast, assets that pay off when the market tanks are valuable. These assets have low betas. Low beta assets have tremendous diversification benefits and are very attractive to hold. Investors, therefore, do not need to be compensated very much for holding them. In fact, if the payoffs of these low beta assets are high enough when the market return is low, investors are willing to pay to hold these assets rather than be paid. That is, assets with low enough betas actually have

[4] A textbook MBA treatment is Bodie, Kane, and Marcus (2011). A more rigorous treatment is Cvitanić and Zapatero (2004).